PuLP 2.5.0

Latest version

```
pip install PuLP
```

Released: about 1 hour ago

PuLP is an LP modeler written in python. PuLP can generate MPS or LP files and call GLPK, COIN CLP/CBC, CPLEX, and GUROBI to solve linear problems.

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GitHub statistics:

Stars: 1,212

Project description

build passing

PuLP is an LP modeler written in Python. PuLP can generate MPS or LP files and call GLPK, COIN-OR CLP/CBC, CPLEX, GUROBI, MOSEK, XPRESS, CHOCO, MIPCL, SCIP to solve linear problems.

Installation

The easiest way to install pulp is via PyPi

If pip is available on your system:

```
python -m pip install pulp
```

Otherwise follow the download instructions on the PyPi page.

Forks: 263

Open issues/PRs: 19

View statistics for this project via Libraries.io, or by using our public dataset on Google BigQuery

Meta

License: BSD License

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Optimization, Linear Programming, Operations Research

Maintainers


js


pchtsp


StuartMitchell

Classifiers

Development Status

- 5 - Production/Stable

Environment

- Console

If you want to install the latest version from github you can run the following:

```
python -m pip install -U git+https://github.com/coin-or/
```

On Linux and OSX systems the tests must be run to make the default solver executable.

```
sudo pulptest
```

Examples

See the examples directory for examples.

PuLP requires Python 2.7 or Python >= 3.4.

The examples use the default solver (CBC). To use other solvers they must be available (installed and accessible). For more information on how to do that, see the guide on configuring solvers.

Documentation

Documentation is found on https://coin-or.github.io/pulp/.

Use LpVariable() to create new variables. To create a variable 0 <= x <= 3:

```
x = LpVariable("x", 0, 3)
```

To create a variable 0 <= y <= 1:

```
y = LpVariable("y", 0, 1)
```

Use LpProblem() to create new problems. Create “myProblem”:

Intended Audience
- Science/Research

License
- OSI Approved :: BSD License

Natural Language
- English

Programming Language
- Python

Topic
- Scientific/Engineering :: Mathematics

slack

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```
prob = LpProblem("myProblem", LpMinimize)
```

Combine variables to create expressions and constraints, then add them to the problem:

```
prob += x + y <= 2
```

If you add an expression (not a constraint), it will become the objective:

```
prob += -4*x + y
```

To solve with the default included solver:

```
status = prob.solve()
```

To use another sovler to solve the problem:

```
status = prob.solve(GLPK(msg = 0))
```

Display the status of the solution:

```
LpStatus[status]
> 'Optimal'
```

You can get the value of the variables using value(). ex:

```
value(x)
> 2.0
```

Exported Classes:

- `LpProblem` – Container class for a Linear programming problem
- `LpVariable` – Variables that are added to constraints in the LP
- `LpConstraint` – A constraint of the general form

> *a1x1+a2x2 …anxn (<=, =, >=) b*

- `LpConstraintVar` – Used to construct a column of the model in column-wise modelling

Exported Functions:

- `value()` – Finds the value of a variable or expression
- `lpSum()` – given a list of the form [a1*x1, a2x2, …, anxn] will construct a linear expression to be used as a constraint or variable
- `lpDot()` –given two lists of the form [a1, a2, …, an] and [x1, x2, …, xn] will construct a linear expression to be used as a constraint or variable

Building the documentation

The PuLP documentation is built with Sphinx. We recommended using a virtual environment to build the documentation locally.

To build, run the following in a terminal window, in the PuLP root directory

```
cd pulp
python -m pip install -r requirements-dev.txt
cd doc
make html
```

A folder named html will be created inside the `build/` directory. The home page for the documentation is `doc/build/html/index.html` which can be opened in a browser.

Comments, bug reports, patches and suggestions are welcome.

- Comments and suggestions: https://github.com/coin-or/pulp/discussions
- Bug reports: https://github.com/coin-or/pulp/issues
- Patches: https://github.com/coin-or/pulp/pulls

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